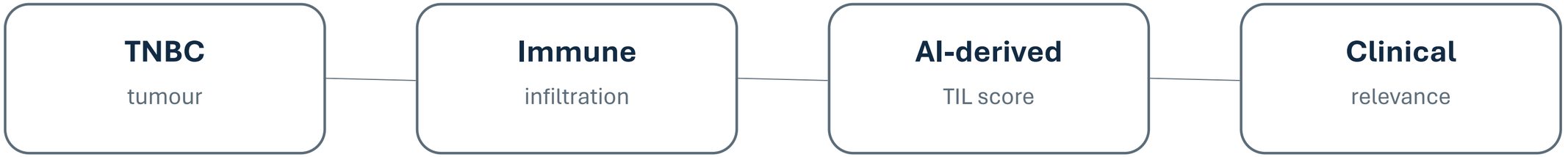
Do immune biomarkers remain valid across populations?

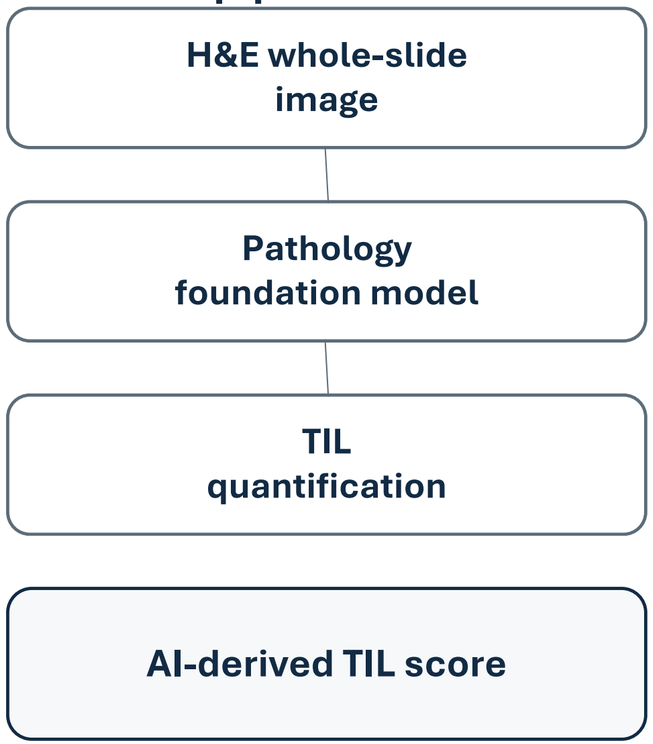


Can we quantify TILs using AI, and more importantly, can that measurement remain trustworthy when we move from one population to another?

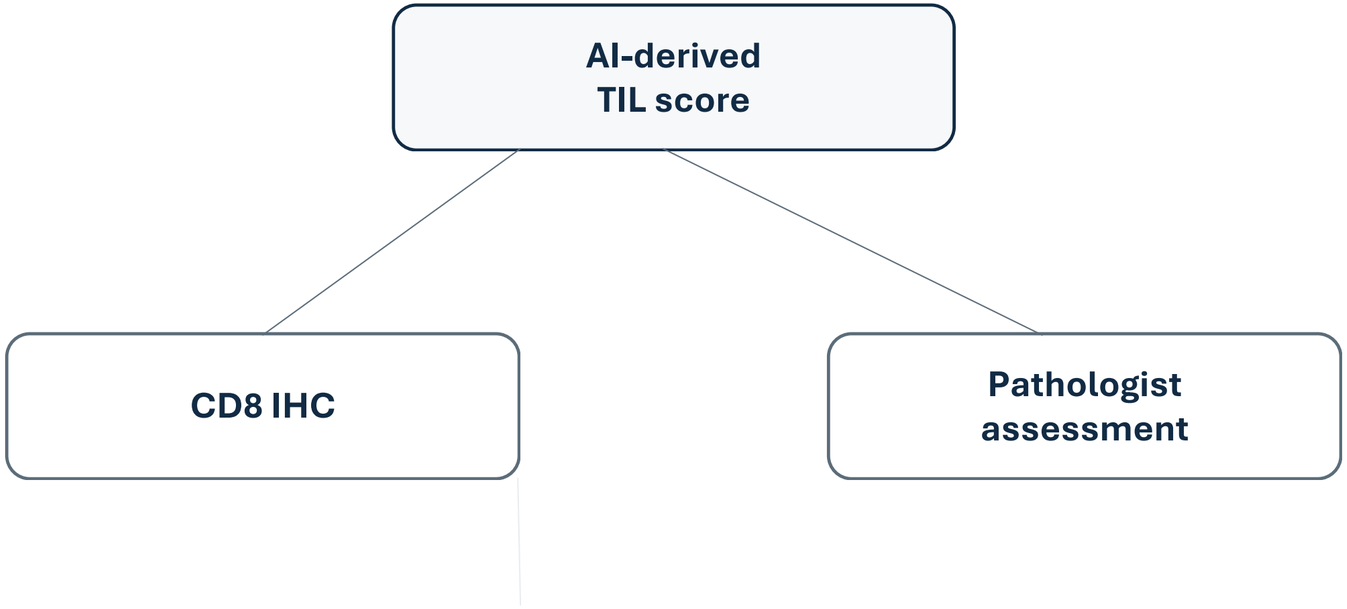
Aya Alaoui Sossée **R&D Data Scientist, Datapathology (Morocco) · PhD Student**

Establish measurement validity before testing generalization

Measurement pipeline



Triangulation against reference measures

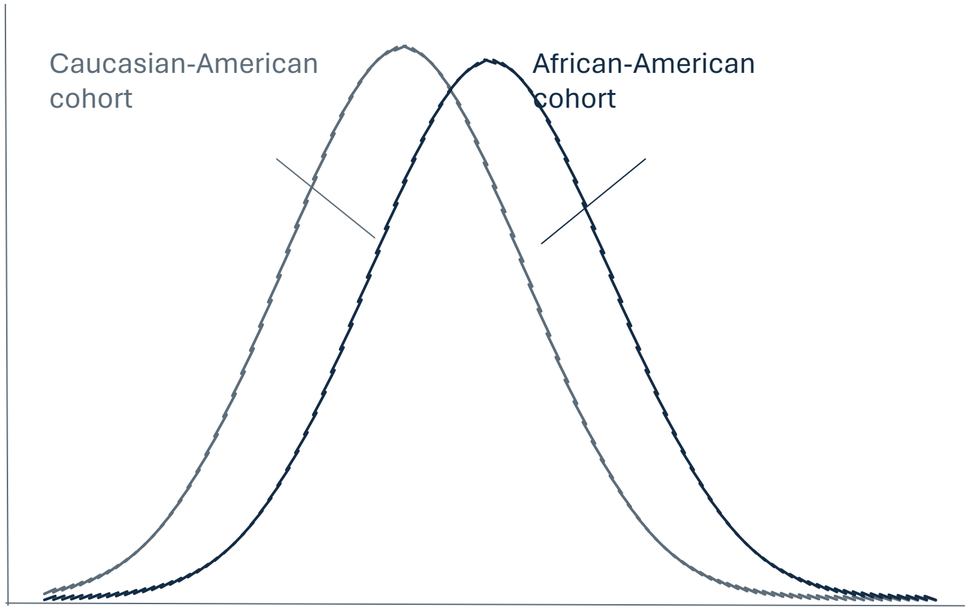


Concordance supports biological validity of the computational measurement.

Test whether TIL measurements differ across populations

Primary comparison

TIL score distribution



Potential confounding

Clinical factors	Age • tumour characteristics Treatment • clinical setting
Technical factors	Tissue processing • staining Scanner type

Adjust for relevant clinical and technical confounders

A distributional difference is not, by itself, evidence of a biological difference.

Move from cross-population validation to a new data-generating environment



Generalization asks whether the validated measurement remains meaningful when acquisition context changes.

A three-stage validation strategy

3 GENERALIZE — Does it transfer to Moroccan data?

2 STRESS-TEST — Does the measurement behave consistently across populations?

1 ESTABLISH — Can we measure TILs reliably?

The purpose is to distinguish biological signal from dataset- and acquisition-specific signal.

From validated measurement to Moroccan TNBC



The endpoint is not a better model.

It is a more trustworthy biological measurement.

Can computational biomarkers remain trustworthy when the population changes?

ACKNOWLEDGEMENT

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Thank you!